Bags, Vectors & Transformers

Topic Models with BERTopic · Bonus, Part 11

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Topic Models: Finding Themes Nobody Defined

LDA (Blei et al., 2003)

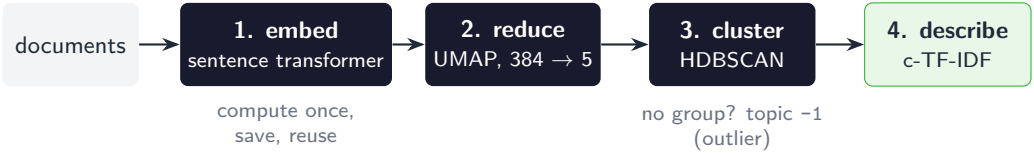
- works on **word counts**
- each document is a **mix** of topics
- you choose the number of topics
- struggles with short texts

BERTopic (Grootendorst, 2022)

- works on **sentence embeddings**
- each document in **one** topic, or none (outlier)
- the number of topics follows from the clustering
- good with short texts: embeddings bring knowledge from pre-training

Thursday's session introduced sentence transformers and document clustering. This module puts them together, and focuses on what decides whether you can trust a topic model: **choices**, **chance**, and **validation**.

BERTopic in Four Steps



c-TF-IDF: glue all documents of a cluster into one "class document", then TF-IDF (Day 1) across classes. Words typical for this cluster, compared to the others, describe the topic.

```
topic_model = BERTopic(embedding_model=embedder,
umap_model=UMAP(n_neighbors=15, n_components=5,
min_dist=0.0, metric="cosine", random_state=42),
hdbscan_model=HDBSCAN(min_cluster_size=20),
vectorizer_model=CountVectorizer(stop_words=STOP))
topics, _ = topic_model.fit_transform(docs,
embeddings)
```

Each step is a separate, replaceable tool. Set them explicitly, so you know (and can report) what the model does.

The Data, and Looking at Topics

2,400 US bill titles

- same source as Day 3 and Bonus 6–8
- eight policy areas, 300 bills each
- plus the Congress (two-year period)
- the model never sees the policy areas: we keep them for **validation**

Domain stop words. "Amend", "act", "provide", "purposes" are in almost every bill title. Left in, they dominate every topic's description.

Read documents, not just top words. A few frequent words can dominate a topic's description while its documents are about something more specific, or a mix.

```
topic_model.get_topic_info()         # sizes, names
topic_model.get_topic(3)             # top words
topic_model.get_representative_docs(3)
topic_model.visualize_barchart()    # interactive
topic_model.set_topic_labels({3:  "..."})
```

Validation: Do Topics Match Anything We Know?

Cross-tabulate topics with the policy areas. A good topic falls mostly in one column; an area may split into several topics.

	Health	**Tax**	**Educ.**	...
topic 0	118	2	4	
topic 1	3	96	1	
topic 2	51	0	30	

Illustration of the layout; the notebook shows the real table.

Summary numbers, from 0 (unrelated) to 1 (identical): **normalized mutual information** (NMI) and the **adjusted Rand index** (ARI). Report the share of outliers left out.

Policy areas are *one* way of dividing the bills, not the "true topics". A model that splits Health into Medicare and public health scores lower, but may be more useful. Use labels to check that topics mean something, and to see how they differ.

Choices and Chance

Your choices shape the topics

`min_cluster_size` is the big one: small values give many small topics, large values a few broad ones. No correct value: it depends on the level of detail your question needs.

Also: the embedding model, UMAP's `n_neighbors`, the stop words.

Outliers: `reduce_outliers` assigns them to the nearest topic. Honest about all documents, but some assignments are forced. Then `update_topics`.

How much is chance?

UMAP is random. A different `random_state` gives a slightly different space, and possibly different clusters.

Fit with several seeds and compare the assignments (ARI between runs, on documents that aren't outliers in either). If topics move, build conclusions only on what holds across runs.

```
for seed in [1, 2, 3]:
m = make_model(seed=seed)
runs[seed], _ = m.fit_transform(docs, embeddings)
```

Topics over Time, and LDA

Topics over time

```
tot = topic_model.topics_over_time(docs, congress)
topic_model.visualize_topics_over_time(tot)
```

Counts depend on how many documents per period are in the sample. Compare **shares per period**, from a sample designed for comparisons over time.

Comparing with LDA

Same number of topics, same stop words, each document's most likely topic. On bill titles (a dozen words) LDA has little co-occurrence to learn from; embeddings already know that *Medicare* and *hospital* are related.

On long documents the gap shrinks, and LDA's mixed membership can be an advantage. For covariates (time, party) in the model itself: the structural topic model (STM, in R).

Labels and Good Practice

Better descriptions

- `KeyBERTInspired`: words closest to the topic's documents
- `MaximalMarginalRelevance`: relevant *and* varied words
- an LLM (Day 4) writes a label from words and documents; check it, a fluent label can be wrong

Good practice

- read documents before naming a topic
- report every choice: embedding model, UMAP, HDBSCAN, vectorizer, seed, outlier handling
- check stability across seeds and settings
- validate: labels, metadata, a hand-coded sample (Bonus 5), experts

Topics are a **lens** for exploring, not a measurement of what documents are "really about". To measure a concept you've defined, a validated classifier (Days 2–4) is usually the better tool.

In the Notebook

- The four steps, set up explicitly, with embeddings computed once
- Inspecting topics: top words, representative documents, your own labels
- Validation against policy areas: cross-table, NMI, ARI
- How `min_cluster_size` changes the model; handling outliers
- Stability across random seeds
- Topics over time, and a comparison with LDA

`bonus/notebooks/11_topic_models_with_bertopic_exercises.ipynb` · 5 exercises · GPU optional